



UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA
DIPARTIMENTO DI FISICA
"GIUSEPPE OCCHIALINI"

CORSO DI LAUREA MAGISTRALE IN FISICA
Anno Accademico 2024/2025

CHARACTERIZATION OF A LANTHANUM BROMIDE CRYSTAL

Andrea Baiocco
Giovanni Locatelli
Gabriele Papasodaro
Alberto Sala

Summary

The experience's aim is to characterize the detection efficiency of a Lanthanum Bromide (LaBr_3) scintillator crystal by studying its intrinsic radioactivity through the decay of ^{138}La . Then starting from this results, we want to identify an unknown radioactive source and estimate the activity of a radioactive source of Cesium (^{137}Cs).

The design of the machine is describe as show in figure 1

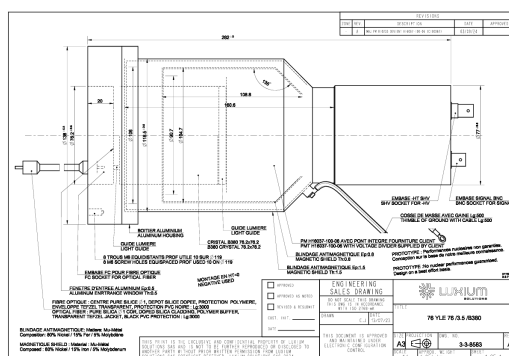


Figure 1: Design instrument

Contents

1	Instruments description	1
1.1	Tools list	1
1.2	PMT and scintillator crystal	1
1.3	Digitizer, AC generator and BNC connections	1
1.4	Support and plates	2
2	Data Analysis	3
2.1	LaBr ₃ scintillator	5
2.1.1	Resolution in Poisson limit	5
2.1.2	Intrinsic radioactivity and Detection efficiency	6
2.2	¹³⁷ Cs source	7
2.2.1	Monte Carlo simulation	7
2.2.2	¹³⁷ Cs measures	8
3	Search for unknown source	9
A	¹³⁷Cesio	11
B	Parameter	12
C	Error propagation	13

Chapter 1

Instruments description

1.1 Tools list

- Phototube with scintillator crystal $\text{La}(\text{: Ce 5\%})\text{Br}_3$
- some radioactive source as ^{137}Cs , ^{22}Na and an unknown source
- voltage generator
- PhotoMultiplier Tube (PMT)
- BNC connections
- digitizer
- plates and supports of different sizes

1.2 PMT and scintillator crystal

The operation of a scintillator crystal is based on the process of radiation-matter interaction and its purpose is to transform the signal given by a photon into a measurable electrical signal. Following the interaction with an incident photon, an electron of the crystal is released, losing energy and exciting the electrons of the semiconductor, taking them from the valence band to the conduction band.

These, de-excited, emit an optical photon that cannot be absorbed by the semiconductor because it is doped. This photon produces, by photoelectric effect on the photocathode, an electron that is amplified inside a photomultiplier by a series of dynodes placed at different potentials, guaranteeing a gain of 10^5 or 10^6 electrons.

In our case, a crystal of Lanthanum Bromide doped at 5% with Cerium was used. Lanthanum Bromide contains a radioactive isotope, ^{138}La (a.i 0.08881%) with a half-life of $1.03 \cdot 10^{11}$ years that decays by electron capture (BR 65.2%) on an excited state of ^{138}Ba . When de-excited, it emits a γ beam at 1436 KeV.

The choice of this crystal was dictated precisely by its intrinsic radioactivity which allows to obtain important information on the detection efficiency of the instrument and allows to carry out its calibration. Furthermore, thanks to the short scintillation time, the crystal allows to have a high counting rate similar to what occurs in plasmas.

1.3 Digitizer, AC generator and BNC connections

About electrical components, in the experience we used an AC voltage generator connected via BNC cables to the instrumentation, thus powering the PMT inside the machine.

Instead, the digitalizer allows to convert the analogical signal we are evaluating into an electrical signal.

1.4 Support and plates

The last component are support and plates of different size. They are used mainly in study of efficiency at various distances. The size of plates are 5 or 10 cm.

Chapter 2

Data Analysis

First, connect the PMT to a high-voltage generator and, since you want to polarize the photo-tube negatively, it must be connected to the negative channel of the generator, while the positive channel of the latter is zeroed via a coaxial cable (which functions as ground).

The photo-tube is then connected to an oscilloscope and, by observing how the output signal varies, the voltage to use is determined. Even at low voltages, a signal begins to appear on the oscilloscope. This is because ^{138}La isotope, being radioactive, emits radiation, which, once converted into "visible" photons by the scintillator crystal, is detected by the PMT. After choosing the potential difference, equal to 1000 V.

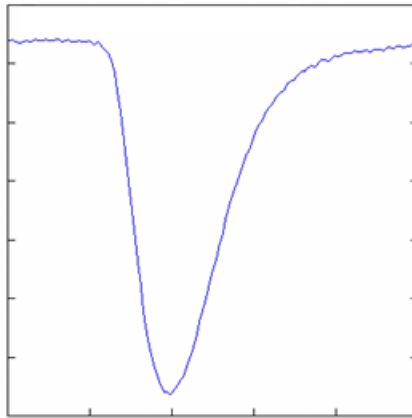
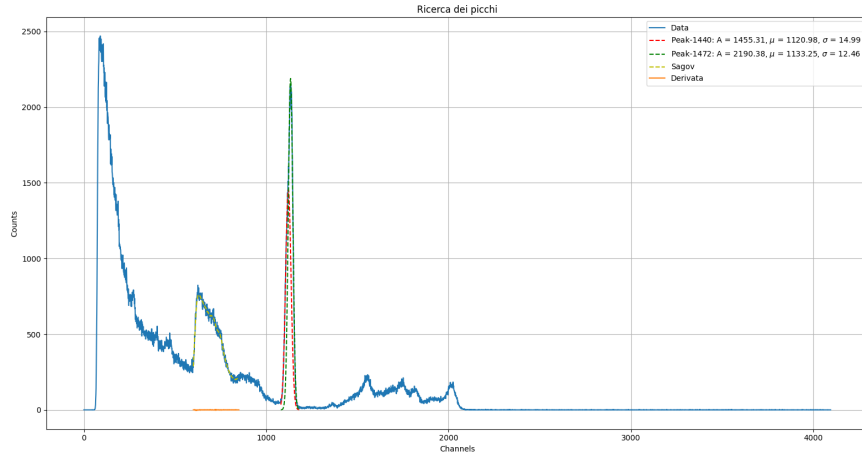


Figure 2.1: Oscilloscope signal

Note that, since the refractive coefficients of the scintillator crystal and the PMT lens differ, it is necessary to insert optical grease (whose coefficient has an intermediate value) between the two surfaces to avoid any total reflection phenomena. Looking at the real-time signal trend output from the photomultiplier tube on the oscilloscope, one notices the presence of some unexpected peaks probably due to the random arrival of a cosmic ray. Therefore, to avoid any type of external influence, the PMT was encapsulated inside the instrument and shielded by a metal cover.

We now proceed to the actual data collection: the output signal is connected to a digitizer that transforms the signal from analog (the one read on the oscilloscope) to digital. To calibrate the digitizer, we identified the three peaks of known energy, respectively: β decay at energy 788,74 keV and the double peaks of ϵ decay at energies 1440 and 1472 keV.



(a) Background

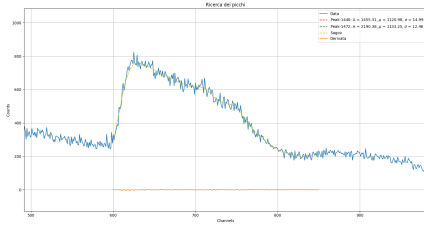
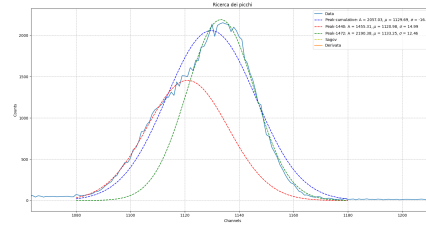
(b) β decay(c) ϵ decay

Figure 2.2: Calibration peaks

Finally performing a linear fit for the conversion from channels to energies, we can evaluate the calibration factor in according to the equation:

$$Energy[keV] = 1,29 \cdot Channel[Ch] + 4,47 \quad (2.1)$$

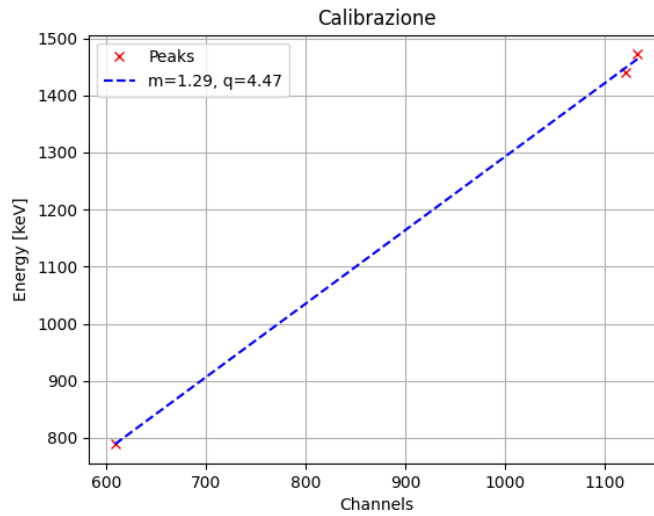


Figure 2.3: Calibration

So applying channels-energies conversion we obtain the follow spectrum (figure 2.4):

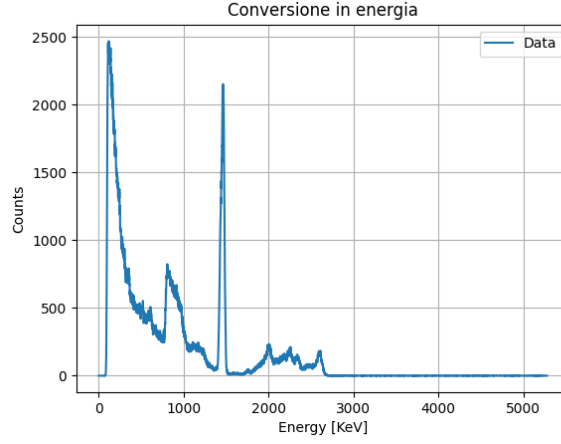


Figure 2.4: Background converted

2.1 LaBr₃ scintillator

2.1.1 Resolution in Poisson limit

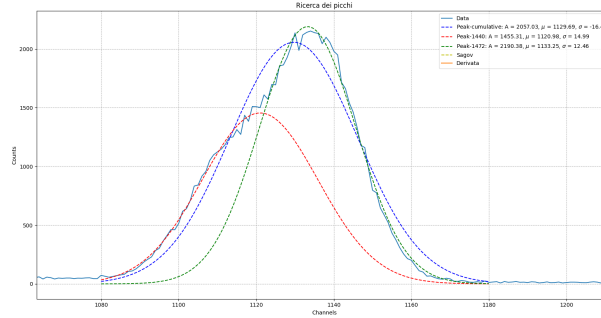


Figure 2.5: Double gaussian fit

The following are the results obtained for the centroids of the two Gaussians and the cumulative one:

- Peak - 1440 keV: (1120.98 ± 14.99) Ch
- Peak - 1472 keV: (1133.25 ± 12.46) Ch
- Cumulative peak: (1129.69 ± 16.43) Ch

At this point, the energy resolution of the detector was estimated:

$$R = \frac{FWHM}{H_0} = \frac{2.35\sqrt{FH_0}}{H_0} \quad (2.2)$$

where FWHM is the width at half-height of the peak and H_0 is its centroid. Taking the centroid of the cumulative Gaussian and its area as a reference.

As previously indicated, we assume that the cumulative peak is the centroid, thus calculating the resolution R of the peak in the Poissonian limit we obtain: 6.99 ± 0.05 %,

instead of given resolution equal to: 3%. This leads us to introduce the Fano factor that is $\approx 18.40 \pm 0.26\%$.

2.1.2 Intrinsic radioactivity and Detection efficiency

The objective is to estimate the absolute intrinsic radiation efficiency of lanthanum, that is, the number of pulses the detector is capable of recording compared to the number of pulses emitted by the source. The procedure is divided into three main steps:

- crystal volume estimation ($V_{crystal}$, equation 2.3)
- source activity estimation (A_{LaBr_3} , equation 2.4)
- estimation of detection efficiency for intrinsic radioactivity ($\epsilon_{abs,RAD.INTR.}$, equation 2.5)

Knowing the nominal efficiency of the instrument, it is possible to estimate the volume of the crystal as the sum of the mass fractions of the various components. Therefore, having a crystal of La(: Ce 5%)Br₃, we can estimate the volume according to the equation:

$$\begin{aligned} V_{crystal} &= \frac{m_{LaBr_3}}{\rho_{LaBr_3}} + \frac{m_{CeBr_3}}{\rho_{CeBr_3}} = \\ &= \left(\frac{\tau_{1/2}}{\ln(2)t} \frac{\#_{LaBr_3}}{BR\epsilon AI} \frac{1}{N_A} \right) \left(\frac{mm_{LaBr_3} 95\%}{\rho_{LaBr_3}} + \frac{mm_{CeBr_3} 5\%}{\rho_{CeBr_3}} \right) = \\ &= 423.07 \pm 67.08 cm^3 \end{aligned} \quad (2.3)$$

The constants and parameter in the formulas are reported in appendix B. The intrinsic activity A_{LaBr_3} is defined as the product of the number of active La atoms and the decay constant λ :

$$\begin{aligned} A_{LaBr_3} &= N_{AA}\lambda = \\ &= N_A \frac{(\rho_{LaBr_3} 95\% + \rho_{CeBr_3} 5\%) V_{crystal}}{(mm_{LaBr_3} 95\% + mm_{CeBr_3} 5\%)} AI \frac{\ln(2)}{\tau_{1/2}} = \\ &= 651.88 \pm 103.15 Bq \end{aligned} \quad (2.4)$$

It remains only to estimate the intrinsic absolute efficiency of the detector:

$$\epsilon_{abs,RAD.INTR.} = \frac{\#_{LaBr_3}}{A_{LaBr_3} \delta t BR} = 18.9870 \pm 0.0002\% \quad (2.5)$$

where $\#_{LaBr_3}$ represents the area of the electron capture peak, that is the number of photons recorded by the detector, δt is the measurement time interval, and BR is the acronym for Branching Ratio, which represents the probability with which ^{138}La decays by electron capture (the remaining percentage is due to decays β^-).

2.2 ^{137}Cs source

2.2.1 Monte Carlo simulation

After calculating the absolute efficiency of the detector relative to the intrinsic radioactivity of the crystal, through a Monte Carlo simulation, we find a relationship between this constant and the efficiency relative to a cesium source. Specifically, we obtain a relationship between the ratio of the two efficiencies and the distance.

$$\epsilon_{ratio} = \frac{\epsilon_{abs,Cs}}{\epsilon_{abs,RAD.INTR.}} \quad (2.6)$$

The parameters were provided by previous studies, the results follow the table:

Distance [cm]	ϵ_{ratio}
5	$1.932 \cdot 10^{-2}$
7	$1.351 \cdot 10^{-2}$
10	$8.671 \cdot 10^{-3}$
12	$6.732 \cdot 10^{-3}$
15	$4.863 \cdot 10^{-3}$
17	$4.022 \cdot 10^{-3}$
20	$3.091 \cdot 10^{-3}$
22	$2.659 \cdot 10^{-3}$
25	$2.148 \cdot 10^{-3}$
30	$1.576 \cdot 10^{-3}$
35	$1.223 \cdot 10^{-3}$
40	$9.517 \cdot 10^{-4}$
45	$7.714 \cdot 10^{-4}$
50	$6.518 \cdot 10^{-4}$

Table 2.1: Geometry efficiency

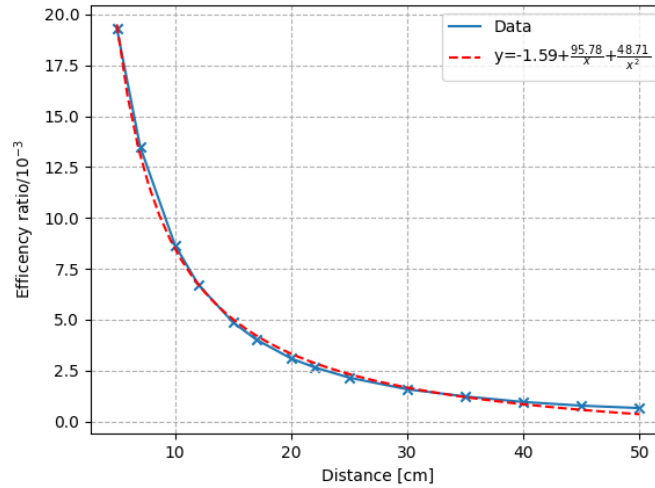


Figure 2.6: Simulation

2.2.2 ^{137}Cs measures

With the data provided it is therefore possible to estimate the activity of the cesium source by analyzing the counts detected in the appendix A and applying the following formula we get the figure 2.7.

$$A_{\text{Cs}} = \frac{\#_{\text{Cs}}}{\epsilon_{\text{abs,Cs}} \delta t} = \frac{\#_{\text{Cs}}}{\epsilon_{\text{ratio}} \epsilon_{\text{abs,RAD.INTR.}} \delta t} \quad (2.7)$$

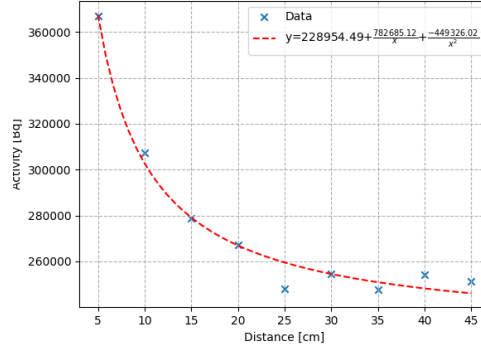


Figure 2.7: Activity

Expected activity: Knowing the initial activity of ^{137}Cs of approximately 41.3 kBq with a half-life of 30.08 years (as reported in the appendix B) the expected activity is estimated using the radioactive decay formula

$$A_{\text{Cs}} = A_0 e^{-\lambda t} = A_0 e^{-\frac{\ln(2)}{\tau_{1/2}} \delta t} = 29.23 \pm 0.1 \text{ kBq} \quad (2.8)$$

we can therefore assume that the data provided for geometric efficiency ϵ_{ratio} are not complete.

Chapter 3

Search for unknown source

During this last part we focused on the research of spectrum analysis for various sources including:

- ^{137}Cs
- ^{22}Na
- unknown source

The purpose of this section is to compare the data obtained for the known sources and compare them with those tabulated on the site <https://www-nds.iaea.org/relnsd/vcharthtml/VChartHTML.html> (radionuclide table), then once the known radionuclides have been confirmed, the unknown source is searched for starting from the energy spectrum that is measured.

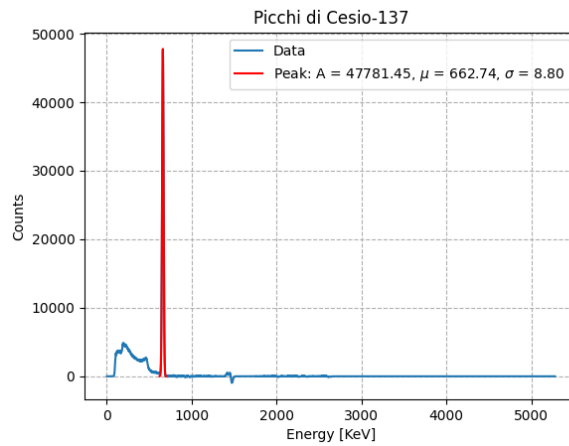
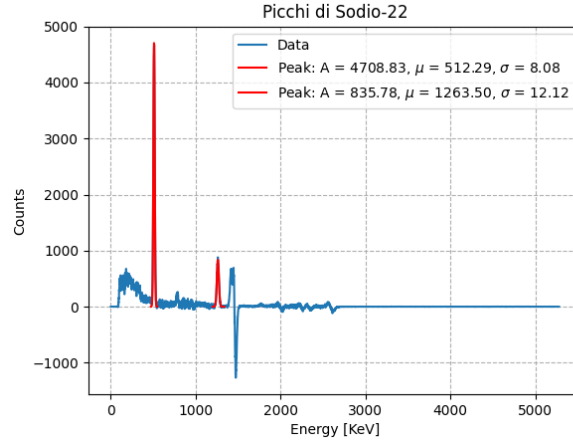


Figure 3.1: ^{137}Cs

Real peak [keV]	Peak [keV]	σ [keV]	Decay Radiation
661.657	662.74	8.80	γ -ray ^{137}Cs

Table 3.1: Energy peak

Figure 3.2: ^{22}Na

Real peak [keV]	Peak [keV]	σ [keV]	Decay Radiation
511	512.29	8.08	e^+e^-
1274.537	1263.5	12.12	γ -ray ^{22}Na

Table 3.2: Energy peak

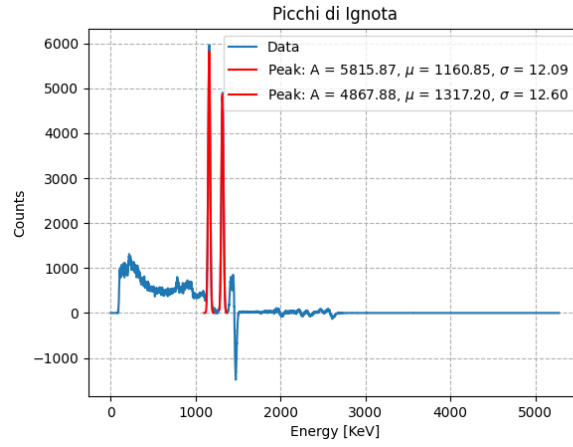


Figure 3.3: Unknown source

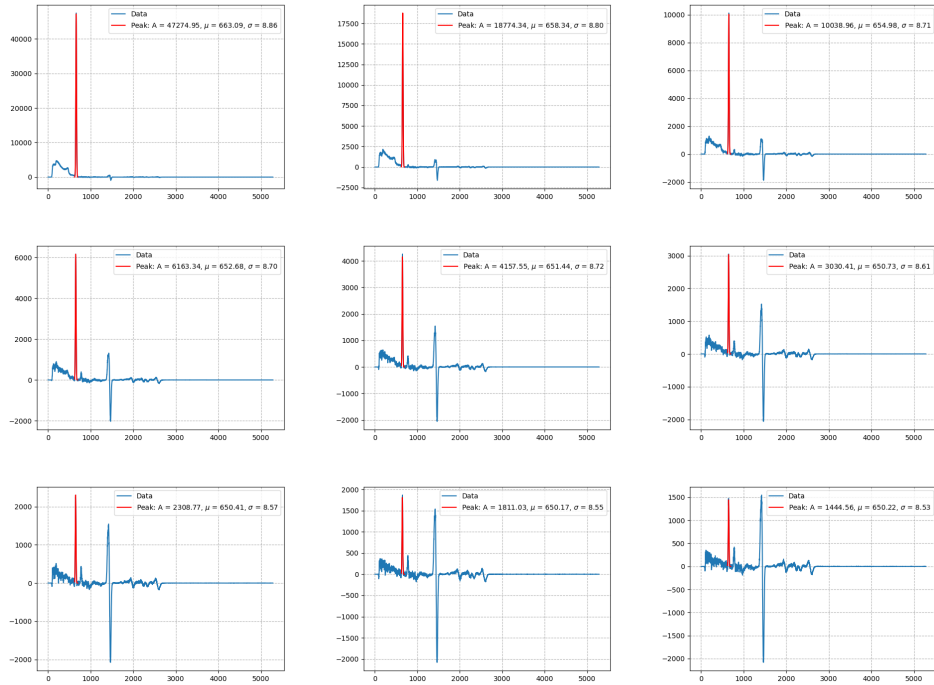
Real peak [keV]	Peak [keV]	σ [keV]	Decay Radiation
1173.228	1160.85	12.09	γ -ray ^{60}Co
1332.492	1317.2	12.6	γ -ray ^{60}Co

Table 3.3: Energy peak

From the analysis we can conclude that the unknown source is ^{60}Co , which is also indicated in the table 3.3. The results obtained are in agreement with what is tabulated.

Appendix A

^{137}Cs



Appendix B

Parameter

^{138}La half-life: $\tau_{1/2} = 1.02 \cdot 10^{11}$ years

^{137}Cs half-life: $\tau_{1/2} = 30.08$ years

Relative efficiency: $\epsilon = 19\%$

Branching ratio: $BR = 65.2\%$

Isotopic abundance: $AI = 0.08881\%$

Avogadro's number: $N_A = 6.023 \cdot 10^{23}$

LaBr_3 molar mass: $mm_{\text{LaBr}_3} = 378.61$

CeBr_3 molar mass: $mm_{\text{CeBr}_3} = 379.82$

LaBr_3 density: $\rho_{\text{LaBr}_3} = 5.06\text{g}/\text{cm}^3$

CeBr_3 density: $\rho_{\text{CeBr}_3} = 5.1\text{g}/\text{cm}^3$

Time: $\delta t = 1200\text{s}$

^{137}Cs activity in 2015: $A_0 = 41.3\text{kBq}$

Appendix C

Error propagation

$$\sigma_F^2 = \sum_{i=1}^N \left(\frac{\delta F}{\delta x_i} \right)^2 \sigma_i^2 + 2 \sum_{i < j} \frac{\delta F}{\delta x_i} \frac{\delta F}{\delta x_j} \rho_{ij} \sigma_i \sigma_j \quad (\text{C.1})$$

σ_i represents the measurement error for the i-quantity, while ρ_{ij} is the correlation coefficient.